Programming Worlds

PROGRAMMING WORLDS ACADEMY

Git & GitHub 101

From Zero to Your First Pull Request

BEYOND CODE. BEYOND LIMITS.

08

MODULE 08

Clone, Fetch and Pull

Get a copy of a repository and keep it in sync with GitHub.

MODULE 08 · LESSON 8.1

Cloning a repository

نسخ مستودع

Learning objectives

- Explain what clone does.
- Clone with HTTPS.
- Choose a folder.
- Verify with git remote -v.

Prerequisites

Module 5 and a GitHub repository.

Introduction

Cloning gives you a full local copy of a repository, including its whole history.

Explanation

`git clone` downloads the repository, creates a folder, and sets up the remote named **origin** for you. You do not need `git init` or `git remote add`.

On GitHub, open the repository, click the green **Code** button, choose **HTTPS**, and copy the URL.

Open Git Bash in the folder where you want the project, for example `Documents`.

Run `git clone URL`. Git creates a folder with the repository name.

`cd my-first-project`, then `git remote -v` to see origin. `git log --oneline` shows the history you received.

In VS Code: **Source Control** → **Clone Repository**, paste the URL, choose a parent folder, and click **Open** when asked.

A private repository asks you to sign in. Git Credential Manager opens a browser window for that.

Real-world analogy

> Cloning is photocopying an entire notebook, including every page ever written, and getting a pointer back to the original.

Diagram

Illustration · Git Bash (not a real screenshot)

```
$ git clone https://github.com/YOUR-USERNAME/my-first-project.git
Cloning into 'my-first-project'...
remote: Enumerating objects: 9, done.
Receiving objects: 100% (9/9), done.
$ cd my-first-project
$ git remote -v
origin  https://github.com/YOUR-USERNAME/my-first-project.git (fetch)
origin  https://github.com/YOUR-USERNAME/my-first-project.git (push)
```

Figure 1 · Illustration: cloning and checking the remote

Commands

```
git clone https://github.com/YOUR-USERNAME/my-first-project.git
cd my-first-project
git remote -v
```

Figure 1 shows the flow. Do not clone inside another repository.

Common mistakes and solutions

Cloning inside a folder that is already a repository.	Pick a normal folder such as Documents.
“Repository not found”	Check the URL and that you are signed in to the right account.
Using the wrong folder name afterwards.	Run `cd` into the cloned folder before other commands.

Practical exercise

> Clone your repository into a new folder next to the original and compare the two.

Expected result

The clone has the same files and `git log --oneline` shows the same commits.

Knowledge check

1. What does clone create?

- ○ A new GitHub account
- ○ A local copy with history and origin
- ○ Only one file

2. Which URL type does this course use?

- ○ HTTPS
- ○ FTP
- ○ Bluetooth

3. Do you need git init after clone?

- ○ Only on Windows
- ○ No
- ○ Yes

Answers

- A local copy with history and origin — It copies everything.
- HTTPS — HTTPS with Credential Manager.
- No — Clone already creates the repository.

Summary

Clone copies a repository, with history, and sets origin for you.

Key vocabulary

clone	Copy a remote repository locally
origin	Default remote name
HTTPS URL	Web address of the repository

Next lesson

Lesson 8.2 — Fetch.

MODULE 08 · LESSON 8.2

Fetching changes

جلب التغييرات

Learning objectives

- Explain fetch.
- Compare local and remote.
- Read “behind” and “ahead”.

Prerequisites

Lesson 8.1.

Introduction

Fetch downloads new work from GitHub without changing your files. It is the safe way to look first.

Explanation

Someone edited the README on GitHub. Your computer does not know yet.

`git fetch` updates your **remote-tracking branches** such as `origin/main`. Your own `main` and your files stay untouched.

`git status` now says your branch is **behind** origin/main by one commit.

Inspect with `git log --oneline main..origin/main` (what you do not have) or `git diff main origin/main`.

“Ahead” means you have commits to push. “Behind” means GitHub has commits to pull. Both at once is “diverged”.

In VS Code, the status bar shows arrows with counts. The Source Control menu (…) has a **Fetch** command.

Real-world analogy

> Fetch is checking the mailbox and reading the letters at the door without bringing them into your room.

Diagram

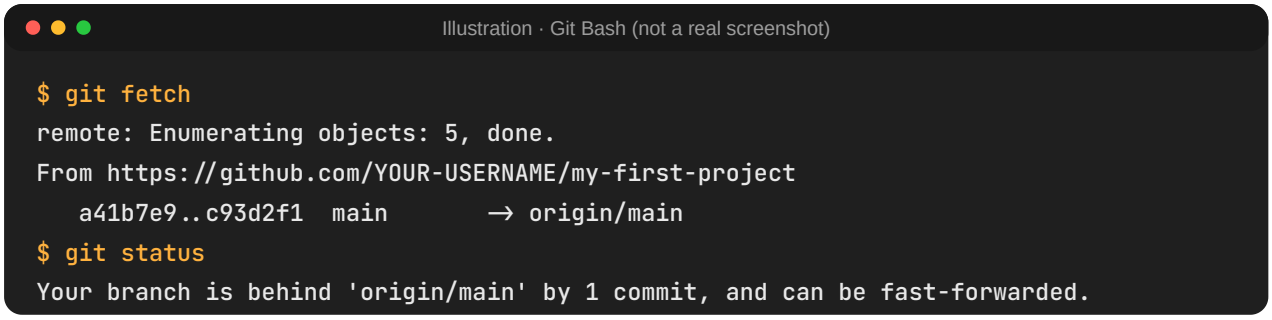


Figure 1 · Illustration: fetch then status

Commands

`git fetch`
`git status`
`git log --oneline main..origin/main`

Figure 1 shows the output after a fetch.

Common mistakes and solutions

Thinking fetch changes your files.	It does not. It only updates origin/*.
Forgetting to fetch before comparing.	Status uses your last known copy of origin.
Panic at “diverged”.	Pull and resolve (Lesson 8.3).

Practical exercise

> Edit README.md on GitHub (the pencil icon), commit there, then fetch locally.

Expected result

`git status` says you are behind by 1 commit and your README is unchanged.

Knowledge check

1. Does fetch change your working files?

- ○ Only on Linux
- ○ No
- ○ Yes, always

2. What does “behind by 1” mean?

- ○ GitHub has one commit you lack
- ○ You have one extra commit
- ○ Git is broken

3. Which branch is updated by fetch?

- ○ feature
- ○ origin/main
- ○ main

Answers

- No — It only updates remote-tracking branches.
- GitHub has one commit you lack — You need to pull.
- origin/main — The remote-tracking branch.

Summary

Fetch looks at GitHub safely. Status then tells you if you are behind or ahead.

Key vocabulary

fetch	Download remote changes without merging
behind	Remote has commits you lack
ahead	You have commits to push

Next lesson

Lesson 8.3 — Pull.

MODULE 08 · LESSON 8.3

Pulling changes

سحب التغييرات

Learning objectives

- Explain pull = fetch + merge.
- Pull safely.
- Handle a pull conflict.

Prerequisites

Lesson 8.2 and Module 7.

Introduction

Pull brings GitHub changes into your branch and files in one step.

Explanation

`git pull` runs `git fetch` and then merges `origin/main` into your current branch.

Commit or stash your work first. A clean tree makes pulls simple.

Run `git pull`. If you are only behind, you get a fast-forward (Figure 1).

If you are ahead too, Git creates a merge commit and may open an editor for the message. Save and close it.

Conflicts are handled exactly as in Lesson 7.3: edit, `git add`, `git commit`.

Good routine: **pull before you start working** and before you push.

In VS Code: **Pull** in the Source Control menu, or **Sync Changes** (pull then push).

Real-world analogy

> Pull is fetching the letters and then filing them into your folder.

Diagram



Figure 1 · Illustration: a fast-forward pull

Commands

`git pull`
`git log --oneline`

Figure 1 shows a fast-forward pull.

Common mistakes and solutions

Pulling with uncommitted edits to the same file.	Commit first.
“fatal: refusing to merge unrelated histories”	You mixed two separate repositories. Re-clone instead of forcing.
Not pulling before push.	Push is rejected; pull first.

Practical exercise

> Pull the README change you made on GitHub.

Expected result

Your README.md now has the new text and the log shows the GitHub commit.

Knowledge check

1. git pull equals…

- ○ add + commit
- ○ fetch + merge
- ○ delete + clone

2. When should you pull?

- ○ Before starting work and before push
- ○ Never
- ○ Only on Fridays

3. What if the pull conflicts?

- ○ Reinstall Git
- ○ Resolve as in a merge
- ○ Delete the repo

Answers

- fetch + merge — It does both steps.
- Before starting work and before push — It avoids rejected pushes.
- Resolve as in a merge — Same process.

Summary

Pull = fetch + merge. Pull often, keep your tree clean.

Key vocabulary

pull	Fetch and merge
Sync Changes	VS Code pull then push
diverged	Both sides have new commits

Next lesson

Module 9 — Collaboration and pull requests.

MODULE 08

Practical lab 8 · Clone, fetch and pull

1. Clone your repository into a new folder: `git clone URL my-first-project-copy` .
2. Open the original on GitHub, edit README.md with the pencil icon, and commit with the message `Update README on GitHub` .
3. In the copy folder run `git fetch` then `git status` . Confirm “behind by 1”.
4. Run `git pull` .
5. Run `git log --oneline` .

Check	Pass if…
Clone	Copy has the same files
Fetch	Status says behind by 1
Pull	Log shows the GitHub commit

> **Evaluation:** 3 of 3 = pass.

Module 8 quiz (pass mark: 70%)

1. What does git clone do?

- ○ Deletes a repo
- ○ Creates an account
- ○ Copies a repository locally

2. Does git fetch change your files?

- ○ Only README
- ○ No
- ○ Yes

3. git pull is…

- ○ fetch + merge
- ○ add + commit
- ○ push + clone

4. “behind by 1” means…

- ○ Nothing
- ○ Remote has a commit you lack
- ○ You must push

5. Which remote name does clone set?

- ○ server
- ○ main
- ○ origin

Answers

- Copies a repository locally — It downloads everything.
- No — It updates origin/*.
- fetch + merge — Two steps in one.
- Remote has a commit you lack — Pull to catch up.
- origin — origin is the default.